

ORF 524 - Fall 2026

Week 2

$$\bullet \quad X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P_{\theta_0} \quad P_{\theta_0} \in \mathcal{P} = \{P_{\theta} : \theta \in \Theta\}$$

$$(1) \quad X_i \sim P_{\theta_0} \quad \forall i \quad \text{and} \quad X_1, X_2, \dots, X_n \text{ are independent}$$

$$P[(X_1, \dots, X_n) \in A] = P[X_1 \in A_1] \cdots P[X_n \in A_n]$$

$$A = A_1 \times \dots \times A_n$$

$$(2) \quad (X_1, \dots, X_n) \sim P_{\theta_0}^{\otimes n} = \underbrace{P_{\theta_0} \times \dots \times P_{\theta_0}}_{n\text{-times}}$$

$$X \sim P_{\theta_0}$$

$$\theta = (\alpha, \sigma)$$

$$p_{\theta_0}(\theta) = \mathbb{E}(P_{\theta_0}) = f(\alpha)$$

$$Y \sim \mathcal{N}(\mu, \sigma^2)$$

$$p(\theta) = \frac{1}{\mu}$$

Why MLE (and the such) work?

$$\hat{\theta}_{MLE} \in \underset{\theta \in \Theta}{\operatorname{argmax}} Q_n(\theta)$$

$$Q_n(\theta) = \frac{1}{n} \log L_n(\theta) = \frac{1}{n} \log f_X(x; \theta)$$

$$= \frac{1}{n} \log \prod_{i=1}^n f_{X_i}(x_i, \theta)$$

$$= \frac{1}{n} \sum_{i=1}^n \log f_{X_i}(x_i, \theta)$$

If

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta}$$

$$\frac{1}{n} \sum_{i=1}^n \log f(x_i, \theta)$$



$$\theta_0 = \arg \max_{\theta \in \Theta}$$

$$\mathbb{E}_{\theta_0} [\log f(x_i, \theta)]$$